COMMUNITY GUIDE

Troubleshooting by symptom

Find the first failing stage, from the camera to the displayed game.

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Start at the first stage that fails: camera, recognition, Controller delivery, RetroPie reception, emulator selection, then the displayed game. Keep a normal gamepad available. Change one setting at a time so you know what fixed the issue.

The **PowerGlove Vision Controller (Arduino UNO Q)** hosts Setup, Glove Academy, and Help. Use **Help -> This console** for addresses specific to your installation. The examples below use placeholders, not addresses that every build shares.

The website will not open

1. Check power and give the Controller time to finish starting.
2. Put your browser device on the same reachable LAN. Check the router's client list for the Controller's current IP address.
3. Try `http://CONTROLLER-IP:8088/setup`. Secure pairing uses `https://CONTROLLER-IP:8443/setup`.
4. If the IP works but `.local` does not, investigate hostname resolution and guest-network/client isolation. With Wi-Fi and USB Ethernet connected, the Controller can have more than one address.

Use the plain `/setup` address; no `?ui=2` suffix is needed. Old query-string bookmarks still open Setup.

Do not change pairing keys to fix an unreachable website. If SSH works, use the configuration troubleshooting reference for application checks. The documented hardware may restart after Shutdown; a disappearing page alone does not prove that power has been removed.

Networking is red or grey

The fourth Setup marker and fourth Off-mode pixel represent a physical **Wi-Fi or Ethernet link**, including supported Ethernet adapters in USB docks.

Marker	Meaning	Next check
Green	At least one detected physical Wi-Fi/Ethernet link is up	Check console-service and authenticated-response markers next
Red	Detected relevant links report disconnected	Check wireless association, Ethernet cable, dock power, and upstream data connection
Grey	The host report is missing, stale, incomplete, or has no recognized interface	Check that the Controller host sampler was installed/upgraded; do not assume the cable is disconnected

Docker bridges and loopback do not make this marker green. A green link does not prove an IP address, Internet access, console reachability, or game delivery. Older Wi-Fi-only telemetry can confirm a connected wireless link but cannot rule out Ethernet when Wi-Fi is down. Use the current Controller installer or upgrade helper to update the host sampler.

The camera view is missing

Gestures off normally closes the camera. Open **Play** or **Glove Academy** and wait for **Starting camera and gesture tracking** to finish. First startup can take longer than switching between active profiles.

If no camera appears, check the powered USB hub, cable, and camera connection. On the Controller host, `lsusb` should show the camera. If it is absent there, the problem is below hand recognition. The documented helper attempts one controlled recovery for the enrolled camera; repeated setting changes will not repair a disconnected USB device. See [Camera selection].

The camera works but the hand is not recognized

Keep one whole hand in frame with its palm facing the camera. Light the hand from the camera side and avoid a bright window behind it. Try a bare hand before changing glove settings; the glove-colour option is only a diagnostic label. Use the first Glove Academy lesson to verify basic detection.

If detection repeatedly disappears, fix visibility before tuning gesture thresholds. Tracking losses and ordinary stationary jitter are different problems and should be reported separately.

The hand is detected but the game does not move

1. Close local Play and Glove Academy; they pause cabinet input. Finish tuning, then explicitly start controller delivery if required.
2. Check the selected player and any request to set a fresh centre. Selecting a player loads their saved centre automatically. Use **Center hand** if no centre is saved or the camera or playing position has changed.
3. Select **Start controller**. Armed means delivery is permitted when a valid game session or intentional manual profile is active; it does not mean packets are always being sent.
4. Check Setup's console-service and authenticated-response markers. A reachable service with unconfirmed authentication suggests pairing needs attention. Neither marker proves emulator input consumption.
5. Confirm that the game has actually started in RetroArch. The exact ROM filename must be registered; `.nes`, `.zip`, and `.7z` are separate entries.
6. Check the emulator and controller selection. For native Super Glove Ball, choose Nestopia (PowerGlove); for its joystick fallback choose FCEUmm.

A filename such as `Gun.Smoke (USA).7z` must keep its punctuation in the registry even though the displayed game name is **Gun Smoke**. See [Register games] and the [Gameplay Guide].

Pairing asks for more than one code

Both pairing methods need the six-digit approval PIN displayed on the Controller matrix and the certificate-ID comparison. **Code pairing** additionally uses the one-time code generated on RetroPie. **Password pairing** additionally uses the RetroPie SSH username and password. The two codes are not interchangeable.

If confirmation expires, select **Start a new confirmation**. The saved console and method stay fixed during the two-minute window; change them after it ends. A failed submitted request also requires fresh confirmation. Save console edits with **Save settings** before pairing. When a submitted pairing attempt finishes, the matrix releases the approval PIN and resumes its normal display. When idle, the glove animation follows your On, Dim, or Off attract setting; active game and status displays still take priority. A completed attempt should not leave the old PIN scrolling for the rest of its two-minute window.

Use the pairing walkthrough; never paste pairing tokens, passwords, or live approval PINs into a public support report.

Movement drifts or feels reversed

Check the selected game profile and centre before changing sensitivity. Hold a relaxed hand at the intended playing position and choose **Center hand**. Support your forearm where practical. Re-centre after moving the camera.

A profile such as Program D intentionally reverses controls. Native Super Glove Ball and joystick-style mappings behave differently, so verify the selected core. If ordinary movement is correct but a gesture is unreliable, use **Glove Academy -> Tune gestures** and describe that symptom to Pixel Pal.

For FCEUmm digital directions, Setup's **Joystick dead zone** adjusts how far the selected player moves before all four directions activate; release remains half that distance. Start with **Use standard size**, then save one small change at a time while watching the live direction indicators. This setting does not change native Super Glove Ball X/Y travel or cure processing latency. Use reach controls under **Glove Academy -> Tune gestures -> Movement reach** for native screen coverage and the latency procedure below for delay. Smaller reach values need less physical hand travel. **Latest coordinate** is the tested default; try **Bounded speed curve** only as a comparison when diagnosing smoothness. Both modes clamp at the saved reach edges. If the Robo-Glove still jumps after the hand leaves and re-enters the picture, confirm that the Controller and RetroPie are on the same current release before changing reach or smoothing. Version 0.4.0 guards one contradictory or unusually distant non-forward reacquisition for one fresh result while allowing strongly aligned forward movement immediately.

Start triggers accidentally, or a gesture stays active

Practise the V sign and its release in Glove Academy. Keep your fingers clearly away from a menu pose while performing another action. Use **A gesture happens accidentally** in Tune gestures if recognition needs personalization.

Menu Guard suppresses D-pad and button output; native continuous positioning still follows the hand. Select **Stop controller** for a dependable pause while repositioning. Do not use extra smoothing to hide a recognition problem.

Controls feel delayed

Close optional camera previews during gameplay. Keep lighting and camera setup stable, then compare deliberate movements and supported stationary holds. Avoid changing several camera, core, and display settings at once.

Software status can locate processing delays but cannot measure the complete hand-to-screen delay. Native Super Glove Ball latency remains an active issue; follow the measurement plan before drawing conclusions from screenshots or timestamps on different computers.

My player or backup looks wrong

Check **Active player** first: player selection applies across browsers. Progress and settings live on the Controller, not in browser storage. Each downloaded backup contains only the selected player's hand setup and excludes Academy progress. Your browser usually puts it in Downloads as `powerglove-hand-setup.json`; keep a separate named copy for each player.

Restore updates the selected player after review. An empty personal-threshold object can simply mean defaults are in use; version-2 backups also carry the effective sensitivity. See backup locations and restore choices.

What to include when asking for help

Record the exact software commit and matrix firmware from the page footer, Controller board variant, camera/dock models, connection type, selected player and profile, core, game filename, and steps that reproduce the symptom. Say whether it occurs in Academy, local Rock Paper Scissors, or only in a cabinet game. Include the four status results and any tracking losses.

Share a short relevant error excerpt or aggregate diagnostic report. Exclude credentials and private video. Raw latency recordings should remain temporary and local unless you explicitly choose to share them. See Contributing for the project's testing and reporting workflow.